



Challenges in Green Energy Adoption: The Role of Digital Sustainability Marketing and Behavioral Processes in Jordan

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Abstract: This paper discusses the effects of the sustainable digital marketing approach on the green purchase intentions (GPIs) in the renewable energy market in Jordan and in particular the behavioral processes that mediate the formation of the environmentally responsible consumption. The quantitative research design was utilized to obtain data based on 237 consumers and processed with the help of the Partial Least Squares Structural Equation Modeling (PLS-SEM), to determine the direct, mediating, and moderating relationships. The results indicate that sustainable digital marketing initiatives have a significant positive impact on the environmental awareness (EA) and GPIs. The EA can be identified as one of the main explanatory factors, and it mediates the connections between sustainable digital marketing and GPIs, which shows that it is the key influential factor of pro-environment consumer behavior. In contrast, the moderating effect of perceived consumer effectiveness (PCE) on the relationship between sustainable digital marketing strategies (SDMS) and GPIs is found to be statistically insignificant. However, the standardized root mean squared residual (SRMR) values of the saturated and estimated models slightly exceeded the recommended 0.08 threshold, suggesting that the structural estimates should be interpreted with caution and that future research should validate the model using larger samples and additional fit indicators. These findings imply that the digital sustainability messages are effective in increasing awareness and behavioral intentions, but the individual perceptions of personal impact might not have a significant effect on strengthening this relationship in the context of the study. On the whole, the research finds that sustainability of digital marketing through the use of eco-friendly practices is a very crucial channel via which sustainable marketing strategies can predict green consumption behavior. The results advance the existing knowledge of the effects of sustainability-focused digital communication on consumer behavior in the developing markets and offer valuable practical recommendations to policymakers and marketers who may want to encourage the use of renewable energy by leveraging the behavioral-based digital communication approaches.

Keywords: Sustainable digital marketing; Green purchase intentions; Environmental awareness; Perceived consumer effectiveness; Renewable energy; Jordan

1. Introduction

The global shift toward sustainable energy systems has increased the importance of consumer behavior in renewable energy markets. Although technological readiness and policy support remain essential for renewable energy adoption, consumer acceptance is still a decisive factor, particularly in emerging economies where sustainability awareness, trust in green technologies, and market readiness may differ from developed contexts (Batool et al., 2024; Obeidat, 2025). In this context, sustainable digital marketing strategies (SDMS) have become important tools for communicating environmental value, building consumer trust, and shaping pro-environmental purchase intentions. SDMS refers to the use of digital platforms and sustainability-oriented content to promote environmentally responsible values, long-term ecological benefits, and green consumption choices (Testa et al., 2015). Unlike traditional promotional strategies, SDMS emphasizes authenticity, environmental education, consumer engagement, and stakeholder-oriented communication. These features make SDMS particularly relevant

to renewable energy markets, where purchase decisions often involve long-term investment, perceived risk, and the need for credible information. Recent studies suggest that sustainability-based digital communication can predict green attitudes and behavioral intentions by improving environmental awareness (EA) and moral engagement (White et al., 2019).

However, despite the growing body of research on green marketing and consumer behavior, the mechanisms through which sustainable digital marketing translates into GPI remain insufficiently explained in renewable energy markets in emerging economies. Prior studies have widely examined the relationship between green marketing, EA, and GPI, but much of this work has been conducted in general consumer-product settings or in contexts where green consumption is relatively more familiar to consumers. Renewable energy decisions are different because they are often associated with higher financial commitment, technical uncertainty, infrastructure dependence, and stronger reliance on institutional and market trust. Therefore, the relationship between SDMS and GPI cannot be assumed to operate in the same way across all green consumption contexts.

Jordan provides a theoretically meaningful setting for examining this issue because renewable energy adoption is not only a matter of individual environmental preference, but also part of a broader national sustainability and energy-security challenge. Jordan has faced long-standing energy dependency, environmental pressure, and the need to diversify energy sources. Although the country has made progress in solar and wind energy, consumer adoption remains uneven, suggesting that informational, psychological, and behavioral barriers continue to affect renewable energy decisions (Alkhawaldeh et al., 2024; Batool et al., 2024). This makes Jordan an appropriate context for testing whether sustainability-oriented digital communication can predict GPI through improved EA, rather than treating the country merely as a new geographical location for an already established model.

EA is widely considered a key psychological mechanism linking sustainability communication with green behavior. Drawing on the Theory of Planned Behavior (TPB) and Value-Belief-Norm (VBN) Theory, EA reflects individuals' knowledge of environmental problems, concern for ecological consequences, and awareness of sustainable alternatives (Ajzen, 1991; Stern, 2000). Environmentally aware consumers are more likely to evaluate renewable energy products positively and to report stronger GPIs (Asif et al., 2023; Paul et al., 2016). Nevertheless, the mediating role of EA in the relationship between SDMS and GPI remains underexamined in renewable energy contexts where consumer decisions require both environmental motivation and confidence in the practical value of green energy solutions. This study therefore treats EA not only as a general predictor of green behavior, but as a specific explanatory mechanism through which digital sustainability communication may shape renewable-energy purchase intention.

Perceived consumer effectiveness (PCE) has also been suggested as an important boundary condition in green consumption research. PCE refers to the extent to which individuals believe that their personal consumption choices can contribute meaningfully to environmental protection (Kumar et al., 2022). Some studies indicate that PCE strengthens the relationship between sustainability messages and green intentions, while other findings suggest that its role may be context-dependent, especially in sectors affected by structural, financial, or infrastructural constraints such as renewable energy adoption (Al Naimat & Liang, 2023; Salah et al., 2023). In Jordan, where renewable energy decisions may depend not only on personal environmental concern but also on affordability, accessibility, policy support, and confidence in service providers, the moderating role of PCE requires further empirical examination.

Based on this discussion, the present study addresses three specific gaps. First, it examines whether SDMS predict GPI in Jordan's renewable energy sector, a context where sustainability communication intersects with energy-security concerns and consumer trust. Second, it investigates EA as a mediating mechanism that explains how SDMS may shape GPI. Third, it examines PCE as a potential moderating factor to determine whether consumers' belief in their personal environmental impact strengthens or weakens this relationship. By doing so, the study contributes to green consumer behavior literature by clarifying the behavioral mechanism and boundary condition of sustainable digital marketing in an emerging renewable energy market. Practically, the study provides insights for renewable energy firms and policymakers seeking to improve consumer engagement, EA, and market acceptance through sustainability-oriented digital communication.

2. Literature Review and Theoretical Framework

This study is mainly grounded in the TPB and the VBN theory. These two perspectives provide the most appropriate theoretical foundation for explaining the proposed relationships among SDMS, EA, PCE, and GPI. TPB explains how consumers form behavioral intentions through attitudes, subjective norms, and perceived behavioral control, while VBN explains how EA, values, and moral responsibility support pro-environmental intentions (Ajzen, 1991; Stern, 2000). Therefore, TPB is used in this study to explain the formation of GPI, whereas VBN is used to explain the awareness-based mechanism through which sustainability communication may be transformed into pro-environmental intention.

2.1 Sustainable Digital Marketing Strategies and Green Purchase Intention

The increasing importance of environmental challenges and the transition toward sustainable energy systems have encouraged scholars and managers to examine how companies can use digital platforms to shape consumer behavior. Within this context, SDMS have become an important approach through which organizations communicate environmental values, present sustainable offerings, and encourage consumers to consider pro-environmental choices (Alghizzawi et al., 2024; Palmié et al., 2025). Unlike traditional digital marketing strategies, which often focus on short-term transactional outcomes, SDMS emphasize long-term environmental value, transparency, ethical engagement, and responsiveness to stakeholders (Leonidou et al., 2013). These strategies may include environmentally oriented content marketing, sustainability communication, interactive storytelling, and personalized messages that connect environmental benefits with consumers' needs and values (Dwivedi et al., 2021). The importance of SDMS becomes clearer in the renewable energy sector because consumer decisions in this field are usually more complex than ordinary green product choices. Renewable energy solutions often involve long-term commitment, financial cost, technical uncertainty, and perceived risk. Consumers may also need to understand technical features, expected benefits, environmental outcomes, and possible economic returns before forming favorable intentions. As a result, credible and accessible communication becomes central to intention formation. In this context, SDMS can function as an informational and interpretive tool that reduces uncertainty, communicates environmental and economic benefits, and increases trust in renewable energy solutions (Batoool et al., 2024). Recent research further suggests that digitally enabled marketing capabilities and customer analytics play a crucial role in enhancing supply chain responsiveness and consumer engagement, thereby strengthening the link between sustainability communication and market acceptance (Alheet & Atieh, 2026).

Stakeholder theory supports this argument by suggesting that firms use sustainability-oriented communication to respond to broader social expectations and strengthen relationships with consumers and other stakeholders (Freeman et al., 2021). Similarly, signaling theory suggests that transparent sustainability communication can reduce information asymmetry and signal a firm's environmental commitment and credibility (Batoool et al., 2024; Palmié et al., 2025; Salah et al., 2023). However, these perspectives are used here only to support the communication role of SDMS. The main explanation of intention formation remains grounded in TPB. GPI is a central outcome in sustainability and consumer behavior research because it reflects a conscious and future-oriented willingness to purchase products or services that reduce environmental harm and support broader sustainability goals (Joshi & Rahman, 2015). In the renewable energy context, GPI goes beyond routine consumption decisions because it may involve adopting solar energy systems, investing in renewable technologies, or choosing alternatives to conventional fossil-based energy sources. Such decisions are generally high-involvement and information-dependent, making them especially responsive to digital communication and sustainability-related information (Paul et al., 2016).

According to TPB, SDMS may predict GPI through three related mechanisms. First, SDMS can shape positive attitudes by highlighting the environmental, economic, and social benefits of renewable energy. Second, digital platforms may reinforce subjective norms by presenting renewable energy adoption as a socially desirable and increasingly accepted behavior. Third, SDMS can enhance perceived behavioral control by reducing information barriers, clarifying product attributes, and providing consumers with practical guidance on renewable energy choices (Ajzen, 1991; Amayreh et al., 2025; Islam et al., 2024). In this way, SDMS are not only promotional tools; they help consumers evaluate renewable energy options more confidently and form stronger GPIs. Previous empirical evidence also indicates that sustainability-based digital communication can improve perceived relevance, trust, emotional engagement, and pro-environmental consumption outcomes (Freeman et al., 2021; Leonidou et al., 2013; Yu, 2025). Digital channels also allow continuous and two-way communication, enabling firms to provide personalized sustainability messages that fit consumer values and preferences (Palmié et al., 2025). This relationship is particularly important in emerging markets such as Jordan, where renewable energy adoption remains developing and consumers' awareness levels may differ across segments. In such contexts, digital marketing channels can become major sources of information that shape consumer knowledge, perceptions, and attitudes toward renewable energy solutions (Al Kurdi et al., 2025; Almrafee & Akaileh, 2024). Well-designed SDMS may therefore reduce informational gaps, improve perceived benefits, and support consumers' intention to shift toward green energy alternatives (Al Naimat & Liang, 2023; Awamleh et al., 2025; Sandri et al., 2020). Moreover, the effectiveness of such digital strategies depends on an organization's digital culture and leadership capabilities, which enable the integration of technological supports into sustainability-oriented communication (Obeidat, 2026). Based on this reasoning, the following hypothesis is proposed:

H1: SDMS have a significant positive effect on consumers' GPIs.

2.2 Sustainable Digital Marketing Strategies and Environmental Awareness

Although SDMS may directly predict GPI, their influence is also expected to operate through consumers' EA. In this study, EA is conceptualized as consumers' awareness of environmental consequences, including their

knowledge of environmental problems, their understanding of the relationship between human activities and ecological degradation, and their recognition that consumption choices can either reduce or intensify environmental harm (Li et al., 2020; Stern, 2000). This conceptualization is consistent with VBN theory, in which awareness of consequences represents a central belief-based mechanism that helps activate personal responsibility and moral obligation toward pro-environmental behavior (Kapoor et al., 2021; Stern, 2000). From a VBN perspective, SDMS can enhance EA by making environmental consequences more visible, understandable, and personally relevant to consumers. Digital platforms allow organizations to communicate sustainability-related information through repeated exposure, visual content, interactive messages, personalized communication, and digital storytelling. These tools can help consumers understand environmental problems, the benefits of renewable energy, and the consequences of unsustainable consumption practices (Dwivedi et al., 2021; Yu, 2025). Unlike traditional one-way communication, digital channels enable consumers to engage continuously with sustainability information and connect it to their own daily consumption decisions (Khuan et al., 2024; Yang et al., 2021).

Accordingly, SDMS are expected to strengthen EA not merely by providing information, but by increasing consumers' awareness of the environmental consequences associated with their choices. This distinction is important because, within VBN theory, EA is not treated as a general knowledge variable only; rather, it represents a belief-based awareness that can support the activation of personal norms and moral responsibility. When consumers become more aware of how renewable energy adoption may contribute to environmental protection, they may be more likely to perceive green consumption as personally and socially meaningful. TPB is used here only as a supporting perspective, not as the primary theoretical foundation. While TPB suggests that greater awareness may contribute to more informed attitudes and perceived ability to evaluate green alternatives (Ajzen, 1991), the main theoretical explanation for the SDMS-EA relationship is grounded in VBN theory. This mechanism is especially relevant in emerging markets such as Jordan, where formal environmental education and public sustainability discourse may still be developing. In such contexts, digital platforms can serve as important channels for communicating environmental consequences, reducing informational gaps, and strengthening consumers' awareness of renewable energy benefits (Al Kurdi et al., 2025; Almrafee & Akaileh, 2024). By presenting renewable energy solutions in a clear, accessible, and contextually meaningful way, SDMS may increase EA and prepare consumers to evaluate green consumption choices through a stronger sense of environmental responsibility (Batool et al., 2024; Salah et al., 2023). Based on this reasoning, the following hypothesis is proposed:

H2: SDMS have a significant positive effect on consumers' EA.

2.3 Environmental Awareness and Green Purchase Intention

EA also plays an important role in transforming environmental knowledge into behavioral intention. Consumers with higher EA are more likely to recognize the seriousness of environmental issues such as climate change and resource depletion, evaluate the environmental consequences of their consumption choices, and consider renewable energy solutions as meaningful alternatives (Kim & Lee, 2023; Li et al., 2020; Salah & Alzghoul, 2024). This is particularly important in renewable energy markets, where purchase decisions are usually complex, long-term, and influenced by consumers' understanding of environmental and practical benefits. The link between EA and GPI is supported by both TPB and VBN. From the TPB perspective, EA helps form positive attitudes toward renewable energy and increases the perceived relevance of green purchasing. Consumers who understand environmental problems are more likely to evaluate renewable energy positively and to express stronger purchase intentions (Ajzen, 1991). From the VBN perspective, EA activates moral concern, environmental values, and personal responsibility, which can encourage intentions toward environmentally responsible choices (Ahmad et al., 2025; Stern, 2000). Therefore, EA functions as both a cognitive and moral driver of GPI. Empirical studies support this reasoning by showing that higher EA is associated with stronger pro-environmental attitudes and greater readiness to adopt environmentally friendly technologies (Asif et al., 2023; Batool et al., 2024; Kim & Lee, 2023; Van Hoang et al., 2025). In the renewable energy context, consumers with higher EA may be more likely to view renewable energy technologies not only as environmentally necessary, but also as personally and socially beneficial. Accordingly, EA is expected to predict GPI toward renewable energy solutions. Based on this reasoning, the following hypothesis is proposed:

H3: EA has a significant positive effect on GPIs toward renewable energy solutions.

2.4 The Mediating Role of Environmental Awareness

The previous sections explain the direct relationships among SDMS, EA, and GPI. This section focuses on the indirect mechanism through which SDMS may predict GPI by strengthening EA. In this study, EA is positioned as a mediator because sustainability-oriented digital communication may not automatically lead to GPI unless consumers first recognize the environmental consequences of their consumption choices. Therefore, EA represents the mechanism through which external sustainability communication is transformed into internal environmental

concern, responsibility, and intention. The mediating role of EA is primarily grounded in VBN theory. According to VBN theory, awareness of consequences is a key belief-based condition that contributes to the activation of personal norms and moral responsibility (Stern, 2000). In the context of this study, SDMS may expose consumers to information about environmental degradation, renewable energy benefits, and the role of individual consumption in supporting sustainability. When this communication increases consumers' awareness of environmental consequences, it may strengthen their sense of personal responsibility and make green purchasing more morally and environmentally meaningful. Thus, EA serves as a bridge between sustainability-oriented digital communication and GPI by linking external environmental messages to consumers' internal norm-based motivation.

This approach avoids treating EA as only a general cognitive antecedent. Instead, EA is understood as a VBN-related awareness of consequences that can support the development of personal norms and pro-environmental motivation. When consumers understand that their purchasing decisions may contribute to environmental protection or harm, they may become more willing to consider renewable energy products and other green alternatives. Accordingly, SDMS may influence GPI both directly and indirectly by first enhancing EA and, through this awareness, strengthening consumers' moral and motivational readiness to engage in green purchasing.

TPB provides a complementary but secondary explanation for this process. From a TPB perspective, EA may also contribute to favorable attitudes toward green products and improve consumers' perceived ability to evaluate sustainable alternatives (Ajzen, 1991). However, the main theoretical logic of the mediation remains anchored in VBN theory, particularly the role of awareness of consequences in activating responsibility and norm-based motivation. This theoretical prioritization helps clarify the function of EA in the model and avoids conceptual overlap between TPB and VBN.

The proposed mediation is also consistent with a cognitive-normative processing view. Digital sustainability content delivered through interactive platforms, visual messages, and continuous engagement can encourage consumers to process environmental information more deeply (Dwivedi et al., 2021; Yu, 2025). Such processing may increase the salience of environmental consequences and make renewable energy solutions more relevant to consumers' personal and social responsibilities (Khuan et al., 2024; Yang et al., 2021). Prior studies similarly suggest that sustainability communication is more likely to influence behavioral outcomes when it first improves environmental knowledge and awareness, which then supports pro-environmental attitudes, responsibility, and intentions (Kim & Lee, 2023; Van Hoang et al., 2025). The mediating role of EA is particularly relevant in Jordan because consumers may differ in their levels of environmental knowledge, exposure to sustainability information, and familiarity with renewable energy solutions. In such a setting, SDMS may need to first reduce informational gaps and strengthen awareness of environmental consequences before they can effectively predict GPI (Al Kurdi et al., 2025; Almrafee & Akaileh, 2024). Based on this theoretical and contextual reasoning, the following hypothesis is proposed:

H4: EA mediates the relationship between SDMS and GPIs.

2.5 The Moderating Role of Perceived Consumer Effectiveness

Beyond the mediating role of EA, individual psychological factors may also shape how consumers respond to sustainability communication. PCE is one such factor. PCE refers to individuals' belief that their personal consumption behavior can contribute meaningfully to environmental protection and sustainability outcomes (Ekebas-Turedi et al., 2021; Kumar et al., 2022; Tarawneh & Alzghoul, 2026). Consumers with high PCE are more likely to believe that their choices, including purchasing renewable energy solutions, can produce real environmental benefits (Batool et al., 2024; Kumar et al., 2022). In the revised theoretical structure, PCE is mainly linked to the perceived behavioral control component of TPB. Perceived behavioral control reflects consumers' belief that they are capable of performing a behavior and that their actions can lead to meaningful outcomes (Ajzen, 1991). In this sense, PCE can strengthen the effect of SDMS on GPI because consumers who believe their choices matter may be more responsive to sustainability messages. When SDMS emphasize the environmental impact of individual decisions, these messages may be more persuasive for consumers with high PCE because they align with their existing sense of personal agency and responsibility.

By contrast, consumers with low PCE may believe that individual action is too limited to contribute to environmental protection. As a result, even if they are exposed to sustainability-oriented digital marketing, they may be less likely to translate such messages into GPI. Prior studies support this logic by showing that individuals with higher PCE are more responsive to sustainability cues, more willing to accept green products, and more likely to engage with environmentally responsible companies (Joshi & Rahman, 2015; White et al., 2019). At the same time, the moderating role of PCE should be treated with caution because renewable energy decisions are also affected by structural, economic, and institutional factors. In renewable energy markets, consumers' intentions may depend not only on whether they believe their personal actions matter, but also on affordability, infrastructure, policy support, and trust in providers (Batool et al., 2024; Salah et al., 2023). Therefore, this study examines PCE as a possible boundary condition rather than assuming that it will always produce a strong moderating effect. Based

on TPB and prior green consumer behavior research, the following hypothesis is proposed:

H5: PCE positively moderates the relationship between SDMS and GPIs.

3. Methodology

The study employed a cross-sectional quantitative research design to empirically examine the relationships among SDMS, EA, GPI, and PCE in the Jordanian renewable energy sector. The quantitative approach was particularly appropriate because the study aimed to test theoretically derived hypotheses and evaluate the structural relationships among the latent constructs. The cross-sectional design enabled the researchers to capture consumers' perceptions and behavioral intentions at a specific point in time, which was consistent with previous studies in the fields of sustainability, digital marketing, and consumer behavior (Hair & Alamer, 2022). In addition, the survey-based methodology facilitated the systematic collection of standardized data from a relatively large sample, thereby supporting statistical generalization and enabling robust estimation of the proposed model through structural equation modeling techniques.

The focus group targeted in this analysis is consumers in Jordan who have been exposed to renewable energy products and/or sustainability-based digital marketing communications in the past. In the system of relevance and validity of answers, the purposive sampling method was used, which enabled the selection of the participants with sufficient awareness and experience concerning the research context. Participants were recruited through online channels that are commonly used by consumers in Jordan, including social media platforms and online communication groups. Before completing the questionnaire, respondents were asked screening questions to confirm that they had previous exposure to renewable energy products or sustainability-related digital marketing content. Only respondents who met these criteria and completed the questionnaire properly were included in the final sample. The collection of data was done by way of an online structured questionnaire, which was sent to popular online platforms within Jordan, such as social media resources and online communication systems. This strategy is relevant to the research topic of digital marketing and will make it possible to target respondents keeping the most popular backgrounds related to digital and likely to be subjects of online content related to sustainability. 237 valid responses were identified and stored to be analyzed later following the process of data screening. During the screening process, incomplete responses and responses that did not meet the study criteria were excluded. The sample size used is regarded as sufficient in terms of Partial Least Squares Structural Equation Modeling (PLS-SEM), since the sample size is a bit bigger than the traditional 10-times rule, and the more modern approach to the subject of statistical power (Hair & Alamer, 2022).

A self-administered online survey was used to collect data within a specific period. Respondents were informed of the purpose of the study beforehand, and were assured of the confidentiality of their identity plus the voluntary nature of the research. All participants were asked to provide informed consent and fill out the questionnaire. In order to maintain the ethical standards, no personally identifiable information was gathered and the respondents were informed that the provided answers would not be used for any purposes other than academic research. Anonymity of the participants was observed during the data collection and analysis process and thus this reduced the chances of bias in response and increased reliability of the data. Also, procedural items were introduced to enhance the quality of the collected data, such as having clear instructions, a simple format of the questionnaire, and including screening questions that would check the knowledge base of the respondents on renewable energy and how digital sustainability would be communicated.

Since all variables were collected from the same respondents through a self-reported questionnaire during a single period, common method bias was also assessed. Procedural remedies were first applied by ensuring anonymity, using clear instructions, and informing respondents that there were no right or wrong answers. In addition, a statistical test was conducted to examine whether common method bias was a serious concern. The results indicated that common method bias was not likely to affect the validity of the findings. Harman's single-factor test was conducted to assess the possibility of common method bias. The results showed that the first factor accounted for 34.72% of the total variance, which is below the recommended 50% threshold. Therefore, common method bias was not considered a serious concern in this study.

To provide a clearer description of the sample, the demographic profile of the respondents is presented in Table 1. The table includes gender, age based on generational groups, education level, monthly income, exposure to sustainable digital marketing, and familiarity with renewable energy products. These categories were included because responses to SDMS, EA, GPIs, and PCE may differ according to demographic background, digital exposure, and previous familiarity with renewable energy solutions.

As shown in Table 1, the sample included a relatively balanced distribution of male and female respondents, with males representing 53.2% and females 46.8% of the sample. The age distribution shows that Millennials formed the largest group at 41.4%, followed by Generation Z at 26.2% and Generation X at 24.1%. This indicates that most respondents belonged to digitally active generations, which is appropriate for a study focusing on SDMS. In terms of education, the majority of respondents held a bachelor's degree, representing 61.6% of the sample, while 22.4% had postgraduate qualifications. This suggests that the respondents had an adequate educational

background to understand issues related to EA and renewable energy. Regarding income, the largest group reported monthly income between 500 and 999 JD, representing 43.5% of the sample. This is relevant because renewable energy purchase intentions may be influenced by financial capacity. In addition, most respondents reported moderate or high exposure to sustainable digital marketing, with 49.8% indicating moderate exposure and 29.5% indicating high exposure. Similarly, most respondents had moderate or high familiarity with renewable energy products. These results support the suitability of the sample, as the respondents had sufficient exposure to digital sustainability communication and renewable energy products to provide meaningful responses to the study variables.

Table 1. Demographic profile of respondents

Variable	Category	Frequency	Percentage
Gender	Male	126	53.2%
	Female	111	46.8%
Age/Generation	Generation Z: 18–26 years	62	26.2%
	Millennials: 27–42 years	98	41.4%
	Generation X: 43–58 years	57	24.1%
	Baby Boomers: 59 years and above	20	8.4%
	Diploma or below	38	16.0%
Education level	Bachelor's degree	146	61.6%
	Postgraduate degree	53	22.4%
Monthly income	Less than 500 JD	71	30.0%
	500-999 JD	103	43.5%
	1000 JD and above	63	26.6%
Exposure to sustainable digital marketing	Low exposure	49	20.7%
	Moderate exposure	118	49.8%
	High exposure	70	29.5%
Familiarity with renewable energy products	Low familiarity	45	19.0%
	Moderate familiarity	121	51.1%
	High familiarity	71	30.0%

4. Result and Findings

4.1 Assessment of the Measurement Model

This section presents the analytical procedures used to assess both the measurement and structural models (Fornell & Larcker, 1981). The first step involved the measurement models where the constructs were assessed to ensure they had satisfactory values for indicator reliability, internal consistency, and convergent validity. After confirming convergence, the next step was to examine the structural model for the relationships hypothesized among the variables of study. The factor loadings of the variables examined are shown in Figure 1, whereas the reliability and validity of each construct are shown in Table 2. As seen in Figure 1, the factor loadings were comprehensively assessed to ensure that each indicator was indeed an accurate representation of the specific construct. The inclusion of items PCE3, PCE6, GPI3, and EA1, which were indicated to have loadings of less than the recommended 0.70, were indicators of poor indicator reliability, and the removal of these items was a decision made to increase the overall measurement accuracy of the model (Fornell & Larcker, 1981). The subsequent measures taken indicated that all the other indicators had acceptable loading values, verifying that the constructs had genuinely been measured with items that were associated with reliable and meaningful constructs (Hair et al., 2011). The results concerning internal consistency reliability and convergent validity were also provided in Table 2. All of the constructs reported Cronbach's alpha and composite reliability (ρ_{α} and $\rho_{\alpha c}$) values to be above the threshold of 0.70 which indicates strong internal consistency. Furthermore, each construct's Average Variance Extracted (AVE) was above the threshold of 0.50 which indicates that the constructs account for the majority of the variance in relation to the indicators aligned with it. All of these findings suggest that the measurement model satisfied the requirements of reliability and convergent validity, which justifies moving on to the evaluation of the structural model (Aboalghanam et al., 2024; Alzghoul, 2025).

In Table 3 we assess discriminant validity using the Fornell-Larcker criterion. Fornell & Larcker (1981) show that the square root of the AVE for each construct should be more than the correlations for that construct with other constructs. In the table, the results supported discriminant validity according to the Fornell-Larcker criterion. The square root of the AVE for each construct exceeded its correlations with all other constructs. Specifically, the diagonal values for EA, GPI, PCE, and SDMS were 0.790, 0.822, 0.751, and 0.772, respectively. Each of these values was greater than the corresponding inter-construct correlations. Therefore, the constructs demonstrated an adequate level of empirical distinctiveness (Ab Hamid et al., 2017).

To test the discriminant validity of the constructs in the measurement model, the heterotrait-monotrait ratio of

correlations (HTMT) was used. HTMT is viewed as a stricter and sensitive criterion than traditional criteria like Fornell-Larcker criterion. The suggested threshold is 0.85 in the strict case and 0.90 in the liberal case. All HTMT values for constructs are under the recommended thresholds, as shown in Table 4, suggesting good discriminant validity. Although the Fornell-Larcker criterion indicated a potential overlap between EA and GPIs, the HTMT results confirmed that discriminant validity is achieved according to the more stringent criterion. The results indicate that EA has acceptable distinctiveness from GPIs (HTMT = 0.731), PCE (HTMT = 0.614), and SDMS (HTMT = 0.639). In the same way, the discriminant validity of GPIs is satisfactory with PCE (HTMT = 0.806) and SDMS (HTMT = 0.812), albeit somewhat higher, but still within acceptable limits.

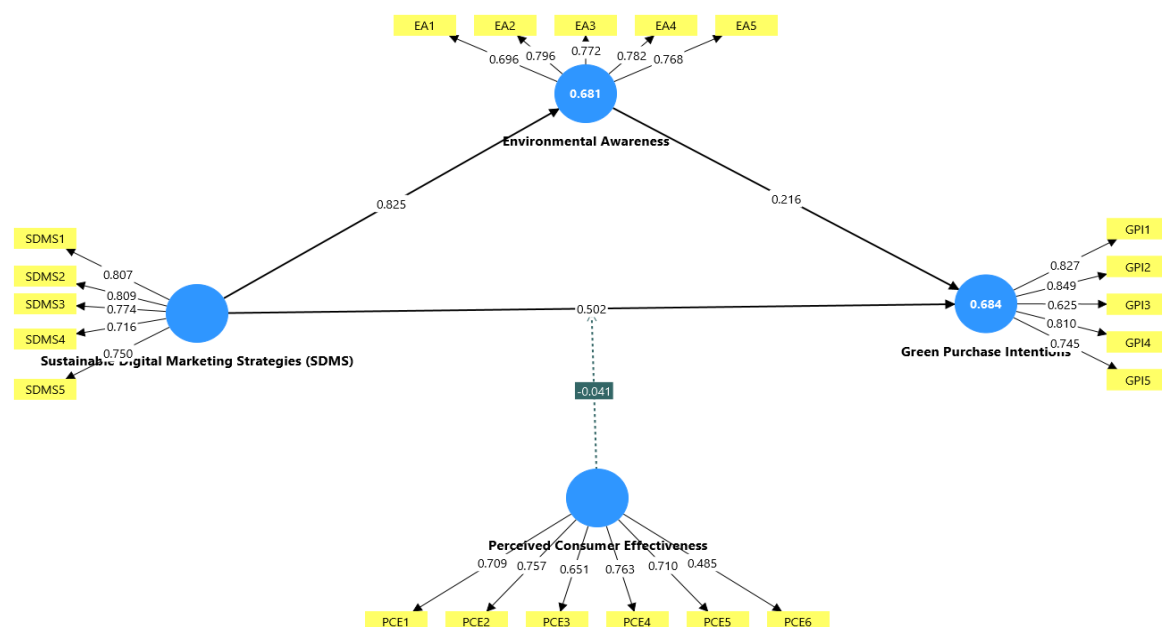


Figure 1. Evaluation of measurement model

Table 2. Construct reliability and validity

Construct	Cronbach's Alpha	Composite Reliability (ρ_a)	Composite Reliability (ρ_c)	Average Variance Extracted (AVE)
Environmental Awareness (EA)	0.799	0.804	0.869	0.625
Green Purchase Intentions (GPIs)	0.839	0.840	0.893	0.676
Perceived Consumer Effectiveness (PCE)	0.743	0.747	0.838	0.564
Sustainable Digital Marketing Strategies (SDMS)	0.830	0.838	0.880	0.596

Table 3. Discriminant validity Fornell-Larcker criterion

Construct	Environmental Awareness (EA)	Green Purchase Intentions (GPIs)	Perceived Consumer Effectiveness (PCE)	Sustainable Digital Marketing Strategies (SDMS)
EA	0.790	—	—	—
GPI	0.706	0.822	—	—
PCE	0.552	0.641	0.751	—
SDMS	0.718	0.735	0.574	0.772

Note: The symbol “—” marks redundant symmetric cells omitted for brevity.

Moreover, the PCE has good discriminant validity with SDMS (HTMT = 0.701). The interaction construct (PCE $\times$ SDMS) is also very low correlated with all other constructs, with the HTMT values ranging from 0.064 to 0.424, thus supporting the empirical distinction of the interaction construct from the main constructs. In general, the HTMT results indicate that all constructs in the model are sufficiently different from each other, thus setting discriminant validity for the measurement model.

Table 4. Heterotrait-Monotrait ratio of correlations (HTMT)

Construct	Environmental Awareness (EA)	Green Purchase Intentions (GPIs)	Perceived Consumer Effectiveness (PCE)	Sustainable Digital Marketing Strategies (SDMS)	PCE × SDMS
EA	—	—	—	—	—
GPI	0.731	—	—	—	—
PCE	0.614	0.806	—	—	—
SDMS	0.639	0.812	0.701	—	—
PCE × SDMS	0.064	0.211	0.424	0.142	—

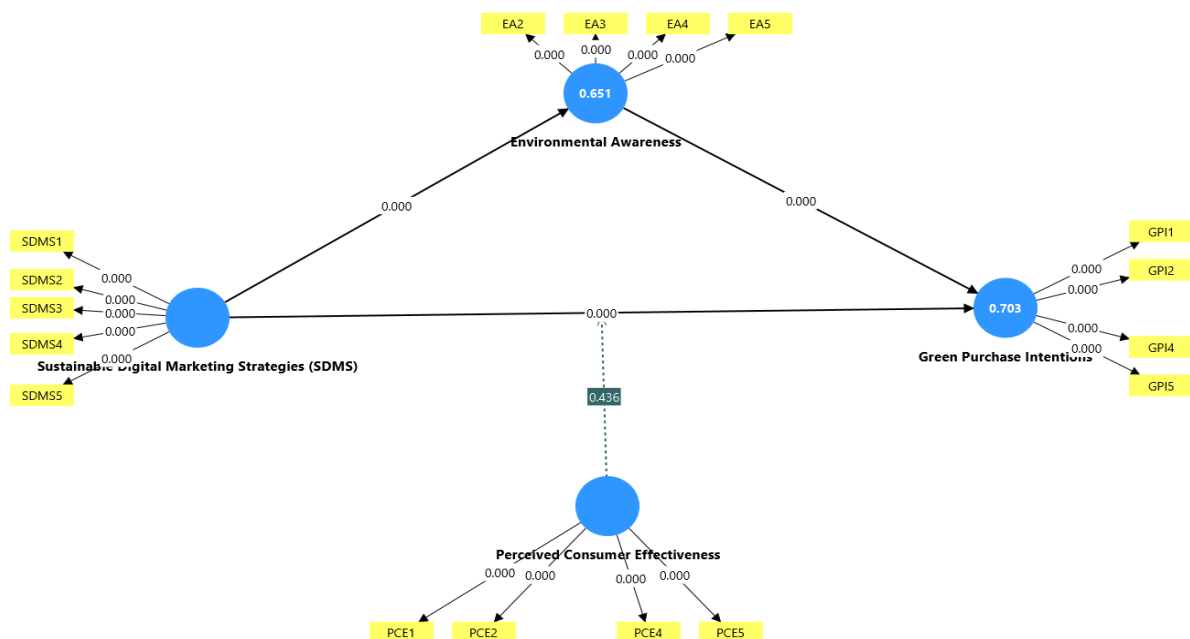
Note: The symbol “—” marks redundant symmetric cells omitted for brevity.

According to the model-fit results presented in Table 5, the standardized root mean squared residual (SRMR) values for the saturated and estimated models were 0.103 and 0.105, respectively. Both values exceeded the commonly recommended threshold of 0.08, indicating that the model did not achieve the preferred level of approximate global fit. Therefore, the SRMR results should be recognized as a limitation rather than interpreted as evidence of good model fit. In PLS-SEM, however, global fit indices are generally treated as supplementary diagnostic measures and should be considered alongside the reliability and validity of the measurement model, the explanatory power of the structural model, the significance of the hypothesized relationships, and the model's predictive relevance. The d_{ULS} and d_G values represented discrepancies between the empirical and model-implied correlation matrices; however, their acceptability could only be determined by comparing them with the corresponding bootstrap-based confidence intervals. Moreover, the normed fit index (NFI) value of 0.616 indicated relatively weak incremental fit and did not independently support a conclusion of acceptable model fit. Accordingly, the analysis proceeded to the assessment of the measurement and structural models, while the elevated SRMR values and limited NFI were explicitly acknowledged as limitations of the model (Fornell & Larcker, 1981).

Table 5. Goodness of fit

Fit Index	Saturated Model	Estimated Model
SRMR	0.103	0.105
d_{ULS}	1.634	1.688
d_G	0.875	0.893
Chi-square	1067.242	1066.861
NFI	0.616	0.616

Note: SRMR = standardized root mean squared residual; NFI = normed fit index.

**Figure 2.** Structural model results

4.2 Evaluation of Structural Model

The structural model results are presented in Table 6 and illustrated in Figure 2. SDMS had a positive and statistically significant direct effect on GPI (H1: $\beta = 0.339$, $t = 5.105$, $p < 0.001$). SDMS also had a strong positive and statistically significant effect on EA (H2: $\beta = 0.807$, $t = 29.020$, $p < 0.001$). In addition, EA had a positive and statistically significant effect on GPI (H3: $\beta = 0.334$, $t = 5.927$, $p < 0.001$). Therefore, H1, H2, and H3 were supported (Kim & Lee, 2023). The indirect effect of SDMS on GPI through EA was also positive and statistically significant (H4: $\beta = 0.270$, $t = 5.838$, $p < 0.001$). Because both the direct and indirect effects were significant, EA partially mediated the relationship between SDMS and GPI. Accordingly, H4 was supported. However, the interaction effect between PCE and SDMS on GPI was not statistically significant (H5: $\beta = -0.024$, $t = 0.779$, $p = 0.436$). This finding indicated that PCE did not significantly moderate the relationship between SDMS and GPI. Therefore, H5 was not supported.

Table 6. Evaluation of structural model

Structural Path	Original Sample (O)	Sample Mean (M)	Standard Deviation	t Statistics (O/STDEV)	p Values
SDMS → GPIs (H1)	0.339	0.340	0.066	5.105	<0.001
SDMS → EA (H2)	0.807	0.808	0.028	29.020	<0.001
EA → GPIs (H3)	0.334	0.335	0.056	5.927	<0.001
SDMS → EA → GPIs (H4)	0.270	0.271	0.046	5.838	<0.001
PCE × SDMS → GPIs (H5)	-0.024	-0.024	0.031	0.779	0.436

Note: EA = environmental awareness; GPIs = green purchase intentions; SDMS = sustainable digital marketing strategies; PCE = perceived consumer effectiveness.

5. Discussion

The present study examined the relationship between SDMS and GPI in the Jordanian renewable energy sector, with EA as a mediating mechanism and PCE as a moderating condition. The findings indicate that SDMS significantly predict GPI, suggesting that sustainability-oriented digital communication can play an important role in shaping consumers' readiness to consider renewable energy solutions. This result is consistent with previous studies showing that green and sustainability-based marketing communication can strengthen consumers' environmental attitudes and intentions (Leonidou et al., 2013; White et al., 2019). However, the present study extends this literature by examining the relationship in the renewable energy sector, where purchase decisions are usually more complex than ordinary green product choices because they involve financial commitment, technical uncertainty, and long-term perceived benefits. The significant relationship between SDMS and GPI also supports the view that digital marketing is not only a promotional tool, but also an informational and educational mechanism. In renewable energy markets, consumers often require credible information before developing purchase intentions. Therefore, digital content that explains the environmental, economic, and social benefits of renewable energy can reduce uncertainty and improve consumer confidence. This finding is in line with prior research suggesting that digital sustainability communication can support pro-environmental behavior by improving knowledge, trust, and engagement (Kapoor et al., 2021; White et al., 2019). At the same time, the result should be interpreted as evidence of association and prediction rather than causality, given the cross-sectional nature of the study.

A major contribution of this study is the confirmed mediating role of EA. The results show that SDMS significantly predict EA, and EA, in turn, predicts GPI. This suggests that sustainable digital marketing may influence consumer intention partly by increasing consumers' understanding of environmental issues and the benefits of renewable energy. This result is consistent with the TPB, which emphasizes the importance of attitudes and perceived understanding in intention formation (Ajzen, 1991), and with the VBN theory, which highlights the role of awareness in activating environmental values and responsibility (Stern et al., 1999). The finding also agrees with recent studies indicating that digital communication can improve environmental knowledge and encourage pro-environmental intentions (Kapoor et al., 2021). The partial mediation result is important because it indicates that SDMS may predict GPI through both direct and indirect pathways. On one hand, digital marketing can directly shape intention by presenting persuasive sustainability messages and making renewable energy more attractive to consumers. On the other hand, it can indirectly shape intention by first increasing EA. This supports the argument that awareness is a key cognitive mechanism through which sustainability communication becomes meaningful for consumers. Compared with studies that focus mainly on the direct effect of green marketing on purchase intention, the present study provides a more detailed explanation of how digital sustainability communication may work in the renewable energy context.

The findings show that PCE did not significantly moderate the relationship between SDMS and GPI; therefore, H5 was not supported. This result should be interpreted strictly as a non-significant moderation finding and should not be taken as evidence that PCE is unimportant or ineffective in the renewable energy context. Although prior

studies have identified PCE as a relevant factor in green consumption behavior (Joshi & Rahman, 2015; Kapoor et al., 2021), the present analysis does not provide statistical support for its moderating role in the relationship between SDMS and GPI. Several methodological explanations may help account for this null finding. First, the PCE scale may not have fully captured consumers' perceived effectiveness in relation to high-involvement renewable energy decisions. Renewable energy adoption differs from routine green purchasing because it often involves higher financial commitment, technical requirements, infrastructure considerations, and longer-term decision processes. Therefore, a general PCE measure may not sufficiently reflect consumers' beliefs about their ability to influence renewable energy transition specifically. Second, the possibility of range restriction should be considered. If respondents' PCE scores were concentrated within a narrow range, the limited variability may have reduced the ability to detect a significant interaction effect. Third, the non-significant result may also be related to statistical power. Since moderation effects are often smaller and more difficult to detect than direct effects, a post-hoc power analysis should be conducted to assess whether the sample size provided sufficient power to detect the observed interaction effect. Accordingly, the unsupported moderation result should be treated with caution. The study only concludes that the data did not support the hypothesized moderating effect of PCE on the SDMS-GPI relationship. Future research should consider using renewable-energy-specific measures of PCE, ensuring adequate variability in PCE responses, and conducting an a priori power analysis to determine the sample size required for detecting interaction effects in similar research contexts. In addition, the SRMR values for the saturated and estimated models slightly exceeded the recommended 0.08 threshold, indicating a model fit limitation that may affect the robustness of the structural estimates and therefore requires cautious interpretation.

From a practical perspective, the findings suggest that renewable energy companies in Jordan should develop digital marketing strategies that are more specific to the realities of the Jordanian renewable energy market. Given Jordan's high solar energy potential and consumers' sensitivity to upfront installation costs, digital campaigns should move beyond general environmental messages and provide practical, decision-support information. For example, companies can use short storytelling videos showing Jordanian households, small businesses, schools, or farms that reduced electricity bills after installing solar energy systems. These videos should present realistic details such as initial cost, financing method, monthly savings, payback period, and challenges faced during installation. Renewable energy firms can also use infographics and short explanatory videos to clarify the steps required to adopt solar energy in Jordan. These may include simple guides on system selection, installation procedures, required documents, connection to the electricity grid, maintenance requirements, and expected savings based on household or business electricity consumption. Interactive calculators can also be used to help consumers estimate installation cost, monthly installment options, expected electricity savings, and payback period. Because upfront cost is a major barrier, digital content should clearly explain available financing options, installment payment plans, bank facilities, and any relevant government or institutional incentive schemes.

In addition, social media campaigns should include practical question-and-answer sessions with renewable energy experts, licensed installers, financing institutions, and consumer protection representatives. Such sessions can address common concerns among Jordanian consumers, including provider reliability, installation quality, warranty terms, maintenance costs, net-metering procedures, and long-term financial benefits. Customer testimonials should also be tailored to local consumer segments, such as apartment owners, villa owners, small retailers, farmers, and SMEs, because each group faces different cost, space, and technical considerations. The findings also suggest that marketing campaigns should emphasize both awareness-building and trust-building. Since EA was a significant mediator, renewable energy firms should explain not only why renewable energy matters environmentally, but also how consumers can practically adopt it in Jordan. For example, digital campaigns could highlight the connection between solar energy adoption, reduced electricity bills, lower dependence on imported energy, and environmental protection. This type of localized content may help consumers connect personal financial benefits with broader sustainability goals.

At the policy level, the results indicate that digital sustainability communication can support public awareness and renewable energy market acceptance when it provides clear and actionable information. Policymakers can collaborate with renewable energy firms, municipalities, universities, banks, and civil society organizations to produce official digital guides on renewable energy adoption. These guides could include step-by-step subsidy or incentive application procedures, lists of approved service providers, financing options, consumer rights, technical standards, and frequently asked questions. Such policy communication should address practical barriers such as cost, financing, technical support, installation procedures, and trust in providers.

6. Conclusion

This study concludes that SDMS are associated with and predict consumers' GPIs in the Jordanian renewable energy sector. However, because the study used a cross-sectional and self-reported survey design, the findings should not be interpreted as evidence of causal effects or actual behavior change. Rather, the results reveal predictive and associational relationships among SDMS, EA, PCE, and GPIs. The findings show that EA serves as a key mediating mechanism, indicating that digital sustainability communication is linked to GPI partly through

consumers' understanding of environmental issues and the benefits of renewable energy solutions. The study also found that PCE did not significantly moderate the relationship between SDMS and GPI. Therefore, this result should be interpreted strictly as a non-significant moderation finding and should not be taken as evidence that PCE is unimportant or ineffective.

The study contributes to the literature on digital marketing, sustainability, and green consumer behavior by explaining how sustainable digital communication is related to attitude-intention processes in an emerging economy. Practically, the findings suggest that renewable energy companies and policymakers in Jordan may develop clear, credible, and educational digital campaigns, including storytelling videos, infographics, customer testimonials, and energy-saving tools, to enhance EA, build trust, and encourage stronger consumer interest in renewable energy adoption. These practical implications should be understood as recommendations for strengthening awareness and purchase intentions, not as evidence that such campaigns directly cause actual renewable energy adoption. It is also important to acknowledge that the SRMR values were slightly above the recommended 0.08 cut-off, indicating that the model fit was not fully optimal; therefore, future studies should confirm these findings with larger samples and a broader set of model fit criteria.

Author Contributions

Conceptualization, A.A. and K.M.A.; methodology, A.A. and K.M.A.; software, A.A.; validation, A.A. and K.M.A.; formal analysis, A.A. and K.M.A.; investigation, A.A.; resources, K.M.A.; data curation, A.A.; writing—original draft preparation, A.A.; writing—review and editing, A.A. and K.M.A.; visualization, A.A.; supervision, K.M.A.; project administration, K.M.A.; funding acquisition, K.M.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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